

A Permutation-Free Vision Transformer for Dense Multi-Emitter Localization

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Abstract

A frame-trained neural network approaches the Bayes estimator of the loss it is trained with, so the choice of output representation and training loss determines which estimator a learned localizer realizes. The coordinate-regression networks studied for dense single-frame emitter localization emit an ordered list of M positions and resolve the unordered nature of the emitter set through a permutation-matched (Hungarian) loss, which makes the target the matched Bayes estimator. Here we study an alternative representation that removes the ordered output altogether. The proposed network, a permutation-free Vision Transformer (ViT-PF), predicts a probability map over a sub-pixel grid and is trained by Kullback–Leibler (KL) divergence against a soft target placed at the true emitter positions; emitter coordinates are read from the map by peak detection. Because the map carries no emitter labels, the formulation is free of the assignment step by construction and cannot exhibit the mean-square (MMSE) degeneracy of unsorted coordinate regression. On the frame model, uniform prior, and localization metric of [preprint submitted], we compare ViT-PF against the coordinate-regression Vision Transformer (ViT) of [preprint submitted] on an in-distribution dataset and against the matched Bayes estimator $\hat{\theta}_B$ computed by quadrature for two emitters. [TBD: one-sentence result summary once re-run — e.g. ViT-PF matches the matched-Bayes optimum to within X nm and equals/exceeds the coordinate-regression ViT across $M = 1\text{--}5$.] The study is a proof of concept on a $700 \times 700 \text{ nm}^2$ field of view.

1. Introduction

Single molecule localization microscopy (SMLM) [1–4] forms a super-resolution image by estimating the positions of individual fluorescent emitters from a sequence of diffraction limited frames. The spatial resolution of the reconstruction and the temporal resolution, defined by the time needed to acquire it, are set against each other by the density of active emitters per frame: a sparse frame carries well separated emitters that are localized easily but requires many frames, whereas a dense frame shortens the acquisition but places several overlapping emitters in a single point spread function (PSF). The estimation of severely overlapping emitters from one frame is

therefore critical to both super spatial and temporal resolution, and it is the problem studied in this paper.

The accuracy of localization has long been analyzed through the Cramér–Rao bound (CRB), the lower bound on the covariance of any unbiased estimator [5–7], which extends to several emitters through the Fisher information matrix of the multi-emitter frame [15–17]. The unbiased Gaussian information achieving estimator for a frame (UGIA-F) realizes the CRB as an oracle on every frame [18, 19], and the global maximum likelihood estimator, approached by multi-start expectation maximization (EM-GML), is a practical benchmark [19–21]. Both benchmarks fail exactly where the emitters overlap: the CRB diverges when two emitters merge, so UGIA-F degrades without limit, while EM-GML is asymptotically efficient yet possesses no optimality property at a finite photon count [19]. The optimum in the mean square sense is neither, but the Bayes estimator that minimizes the risk over a prior of emitter configurations [22]; it carries a bias and is therefore free of the constraint that makes the CRB diverge, so it stays accurate where the unbiased benchmark does not.

It was recently shown [preprint submitted] that a neural network trained by minimizing an average loss over frames drawn from a prior approaches the Bayes estimator of that loss, and that training with a permutation-matched loss makes the target the *matched* Bayes estimator, whose risk is the reported localization metric. Five networks spanning learnable activations, Kolmogorov–Arnold edges, convolution, and self attention were shown to approach EM-GML and to outperform the unbiased oracle UGIA-F in the overlapped configurations where it diverges. In that framework every network, including the Vision Transformer (ViT), estimates an ordered vector of $2M$ numbers read as M positions, and the matched loss, evaluated by the Hungarian assignment, frees the network to produce the M positions in any order.

The matched loss resolves the unordered nature of the emitter set at the level of the *training objective*. A second route is to remove the ordered output at the level of the *representation*: rather than regress M labeled coordinates, a network can predict a spatial distribution over the frame from which the positions are read as peaks. The output is then a probability map rather than a coordinate list, the training loss is a divergence between distributions rather than a matched squared error, and no assignment is performed at any stage. This representation is standard in dense-emitter localizers such as Deep-STORM [23] and DECODE [24], but its relation to the Bayes estimator framework and its comparison, under a common budget, with the coordinate-regression network of the same backbone have not been established.

The contribution of this paper is twofold. First, we define a permutation-free Vision Transformer (ViT-PF) that shares the encoder of the coordinate-regression ViT but replaces the ordered $2M$ -coordinate head with a softmax over a sub-pixel grid, and is trained by KL divergence against a soft

target placed at the true positions. Because the predicted map carries no emitter labels, the formulation is free of the assignment step by construction, and it cannot exhibit the MMSE degeneracy that collapses unsorted coordinate regression to the centroid [preprint, Prop. 1]. Second, we compare ViT-PF with the coordinate-regression ViT of [preprint] on an equal training budget, under the identical frame model, uniform prior, and localization metric, and we measure both against the matched Bayes estimator $\hat{\boldsymbol{\theta}}_B$ computed by quadrature for two emitters. The characterization of which Bayes estimator a KL-trained network approaches is deferred to a companion analysis; here the comparison is empirical. The study is a proof of concept on a field of view (FOV) of $700 \times 700 \text{ nm}^2$.

The paper is organized as follows. Section 2 states the frame model and the Bayes risk framework, the matched Bayes estimator and the localization metric, and defines the sub-pixel classification representation and the KL loss. Section 3 defines the coordinate-regression and permutation-free network estimators and their common training. Section 4 computes the matched Bayes estimator by quadrature for two emitters so that both networks can be measured against the optimum. Section 5 reports the simulation results [TBD]; Section 6 discusses and Section 7 concludes.

2. Estimators and the Bayes risk

2.1 Frame model

We adopt the single-frame model of [preprint submitted]. The mean of the Gaussian readout does not affect the Fisher information and a Gaussian variable whose mean equals its variance is well approximated by a Poisson variable and conversely [15]; the readout mean is therefore set equal to its variance, so that the mixed noise is Poisson and $b(\mathbf{k})$ denotes the combined background and readout density, which varies from pixel to pixel to model the slowly varying autofluorescence and the pixel-dependent readout of an sCMOS camera [28]. The pixel value is the sum of the signal and the noise,

$$V(\mathbf{k}) = S(\mathbf{k}) + B(\mathbf{k}) \sim \text{Poisson}(v(\mathbf{k})), \quad v(\mathbf{k}) = s(\mathbf{k}) + b(\mathbf{k}), \quad (1)$$

where $S(\mathbf{k})$ and $B(\mathbf{k})$ are mutually independent and the pixel values $V(\mathbf{k})$ are independent across the pixels $\mathbf{k} \in \Omega$. For M emitters at positions $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_M)$ the signal mean is $s(\mathbf{k}) = \sum_{m=1}^M s(\mathbf{k}; \boldsymbol{\theta}_m)$, where $s(\mathbf{k}; \boldsymbol{\theta}_m)$ is the mean photon count contributed to pixel $\mathbf{k}$ by an emitter at $\boldsymbol{\theta}_m$ through the Gaussian PSF of width $\sigma = 1.3238/\alpha$, $\alpha = 2\pi \text{NA}/\lambda$ [1]. The data frame is denoted V and its probability mass function, which is also the likelihood of $\boldsymbol{\theta}$, is

$$f_V(V; \boldsymbol{\theta}) = \prod_{\mathbf{k} \in \Omega} \frac{v(\mathbf{k})^{V(\mathbf{k})} e^{-v(\mathbf{k})}}{V(\mathbf{k})!}. \quad (2)$$

Dropping the constant independent of $\boldsymbol{\theta}$, the per-frame log-likelihood is

$$\ell_1(\boldsymbol{\theta}) = \sum_{\mathbf{k} \in \Omega} [V(\mathbf{k}) \ln v(\mathbf{k}) - v(\mathbf{k})], \quad (3)$$

and the Fisher information matrix [15, 19] is

$$\mathbf{F}(\boldsymbol{\theta}) = \sum_{\mathbf{k} \in \Omega} \frac{1}{v(\mathbf{k})} \frac{\partial v(\mathbf{k})}{\partial \boldsymbol{\theta}} \frac{\partial v(\mathbf{k})}{\partial \boldsymbol{\theta}^T}. \quad (4)$$

An estimator estimates $\boldsymbol{\theta}$ from the single frame V , that is, an estimator is a function $\hat{\boldsymbol{\theta}}(V)$.

2.2 Bayes risk

The estimators compared in this paper rest on a common criterion, the Bayes risk, the average estimation error over both the noise in the frame and the emitter positions that produced it [preprint, 22]. Let $p(\boldsymbol{\theta})$ denote the prior from which the emitter positions are drawn and $L(\hat{\boldsymbol{\theta}}, \boldsymbol{\theta})$ a loss. The risk of an estimator is

$$R(\hat{\boldsymbol{\theta}}) = \iint L(\hat{\boldsymbol{\theta}}(V), \boldsymbol{\theta}) f_V(V; \boldsymbol{\theta}) p(\boldsymbol{\theta}) dV d\boldsymbol{\theta}, \quad (5)$$

and the Bayes estimator is the minimizer of R over all functions $\hat{\boldsymbol{\theta}}(\cdot)$ for the loss L and prior p ; its risk is the smallest attainable by any estimator, so it is the bound against which every other estimator is measured. The prior $p(\boldsymbol{\theta})$ is uniform on the central 3×3 pixel block $\mathcal{B}$ of the FOV, the maximum entropy choice on a bounded support that commits to no particular arrangement of the emitters; under this prior the posterior $p(\boldsymbol{\theta} \mid V)$ is proportional to the likelihood (2) restricted to $\mathcal{B}$ [preprint].

2.3 The matched Bayes estimator and the localization metric

The loss used throughout is the matched squared error L_M , evaluated by the Hungarian assignment between the estimated and the true positions, which removes the labels before the

error is formed [preprint]. Its Bayes estimator is the matched Bayes estimator $\hat{\boldsymbol{\theta}}_B$, whose risk is exactly the localization metric reported here, the average root-mean-square matched distance, ARMSMD-H. For an estimate $\hat{\boldsymbol{\theta}}$ and the truth $\boldsymbol{\theta}$ of M emitters the matched distance is

$$\text{MD}_H(\hat{\boldsymbol{\theta}}, \boldsymbol{\theta}) = \left(\min_{\pi \in S_M} \frac{1}{M} \sum_{m=1}^M \|\hat{\boldsymbol{\theta}}_m - \boldsymbol{\theta}_{\pi(m)}\|^2 \right)^{1/2}, \quad (6)$$

and ARMSMD-H is the average of MD_H over the frames of a dataset [preprint; cf. 5]. In contrast, the unsorted squared error L_{SE} has as its Bayes estimator the posterior mean of each emitter index, the MMSE estimator, whose two estimates coincide at the posterior centroid for exchangeable labels and never resolve the emitters [preprint, Prop. 1]. This degeneracy is the reason squared-error coordinate regression fails at overlap, and it motivates the representation of the next subsection, which removes the labels rather than repairing them by matching.

2.4 The sub-pixel classification representation

The estimators above return a labeled or matched list of M coordinates. We now define a representation that returns a spatial distribution instead. The FOV is covered by a grid of $S_c = J_x J_y$ sub-pixel cells of spacing $\Delta_c = \Delta_x / r$, where the integer r is the up-sampling factor and $\Delta_x = \Delta_y$ is the pixel size; cell j is centered at the node $\boldsymbol{\zeta}_j$. An estimator in this representation returns a probability vector over the grid,

$$\hat{\mathbf{P}} = (\hat{P}_1, \dots, \hat{P}_{S_c}), \quad \hat{P}_j \geq 0, \quad \sum_{j=1}^{S_c} \hat{P}_j = 1, \quad (7)$$

that is, a point on the simplex Δ^{S_c-1} , read as a density of emitters over the FOV. The target for a configuration $\boldsymbol{\theta}$ is the soft map obtained by placing a normalized Gaussian bump of width one sub-pixel at each true position and summing over the emitters,

$$P_j^*(\boldsymbol{\theta}) = \frac{1}{Z} \sum_{m=1}^M \exp\left(-\frac{\|\boldsymbol{\zeta}_j - \boldsymbol{\theta}_m\|^2}{2\Delta_c^2}\right), \quad Z = \sum_{j=1}^{S_c} \sum_{m=1}^M \exp\left(-\frac{\|\boldsymbol{\zeta}_j - \boldsymbol{\theta}_m\|^2}{2\Delta_c^2}\right), \quad (8)$$

so that $P^*(\boldsymbol{\theta})$ is a distribution on the grid. The soft bump gives partial credit to a near node and encodes the sub-pixel uncertainty of a position; the hard limit places unit mass on the nearest

node. The discrepancy between a predicted map $\hat{\mathbf{P}}$ and the target is measured by the Kullback–Leibler divergence,

$$L_{\text{KL}}(\hat{\mathbf{P}}, P^*) = \sum_{j=1}^{S_c} P_j^* \ln \frac{P_j^*}{\hat{P}_j}, \quad (9)$$

which equals the cross entropy of $\hat{\mathbf{P}}$ relative to P^* up to the constant entropy of P^* , so minimizing (9) is minimizing the cross-entropy classification loss over the grid. The target (8) is formed from the emitter *set* and carries no labels; the loss (9) contains no assignment between predicted and true emitters. The M coordinates are recovered from $\hat{\mathbf{P}}$ at test time by detecting its M dominant peaks with non-maximum suppression (NMS) over a radius below the smallest inter-emitter separation of interest, each reported position being the center of its peak cell. For a fixed configuration observed over several frames the maps may be averaged before peak detection, which suppresses the per-frame noise. The grid imposes a quantization error of standard deviation $\Delta_c/\sqrt{12}$ per axis on the reported centers; this is made small by the choice of r in Section 4. Which Bayes estimator the loss (9) induces, in the sense of Section 3.1, is outside the scope of this proof of concept and is deferred.

3. Neural network estimators

3.1 What a frame-trained network estimates

A neural network estimator is a function of the frame parameterized by the network weights $\mathbf{w}$. The weights minimize the empirical average of a loss L over a set of training frames,

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \frac{1}{N_t} \sum_{n=1}^{N_t} L(g_{\mathbf{w}}(V_n), \boldsymbol{\theta}_n), \quad (10)$$

where the N_t training pairs $(\boldsymbol{\theta}_n, V_n)$ are generated by drawing $\boldsymbol{\theta}_n$ from the prior $p(\boldsymbol{\theta})$ and then V_n from the frame model (1), and $g_{\mathbf{w}}(V)$ is the network output in the representation of the head. Equation (10) is a sample version of the risk (5) under the loss L ; as N_t grows the sample average converges to R , and the minimizer over all functions of the frame is the Bayes estimator of that loss [preprint]. A frame-trained network therefore approximates the Bayes estimator of whichever loss it is trained with, departing from it only through the finite capacity of the network, the finite number of training frames, and the optimization. For the coordinate-regression head and the matched loss L_M the target is the matched Bayes estimator $\hat{\boldsymbol{\theta}}_B$; for the classification head and the

KL loss L_{KL} the target is the Bayes estimator of the KL risk in the map representation, whose characterization is deferred as noted in Section 2.4. In either case the network is built with the system parameters through its training frames and additionally uses the prior, so the comparisons of this paper are made on frames drawn from $p(\theta)$.

3.2 Network architectures

Both networks share a single backbone and differ only in the output head and the training loss, so that any difference in their performance is attributable to the representation rather than to the capacity. The backbone is the Vision Transformer [35]: each pixel of the frame is treated as one token, a learnable class token is prepended, learnable positional embeddings are added, and a stack of transformer encoder blocks with multi-head self attention is applied. The input is the frame as a vector of $K_x K_y$ pixel values normalized by the largest value of the frame, so the network is given the shape of the frame rather than its photon count.

Coordinate-regression head (ViT). The class token output is mapped by a multilayer perceptron to $2M$ numbers read as M positions in nanometers [preprint]. The network is trained with the matched loss L_{M} evaluated by the Hungarian assignment, so no ordering of the estimates is imposed. This is the baseline and is the ViT of [preprint].

Permutation-free head (ViT-PF). The outputs of all encoder tokens are concatenated and mapped by a multilayer perceptron and a softmax to the probability map $\hat{\mathbf{P}}$ of (7) over the $S_c = J_x J_y$ sub-pixel grid. The network is trained with the KL loss L_{KL} of (9) against the soft target (8), and the M positions are read from the map by peak detection with NMS as in Section 2.4. The head uses all tokens rather than the class token alone because the map is a dense spatial output, whereas the coordinate head reads a global summary from the class token.

3.3 Training

Both networks are trained by minimizing (10), the ViT with the matched loss L_{M} and ViT-PF with the KL loss L_{KL} . Training frames are generated on the fly: a fresh batch of emitter configurations is drawn from the prior and a fresh batch of frames is generated at every step, so no training frame is presented twice and N_t equals the total number of frames generated. The optimizer is Adam [36] with a cosine annealing learning rate [37]. Both networks are trained with the same batch size, the same number of steps, and therefore the same number of training frames, so that their accuracies and convergence are compared on an equal budget. A separate network is trained for each emitter number M .

4. Performance of the matched Bayes estimator for two emitters

Both networks are measured against the matched Bayes estimator $\hat{\theta}_B$, which has no closed form for $M \geq 2$ and must be computed before any estimator can be judged. Following [preprint], we consider $M = 2$, where the posterior can be integrated numerically by quadrature. This section states the computation; the validation that no estimator falls below $\hat{\theta}_B$ on frames drawn from the prior is reported in Section 5.

4.1 The posterior on a grid

The prior $p(\theta)$ is uniform on the central 3×3 pixel block $\mathcal{B}$, both emitters drawn independently and uniformly from that region, so the posterior is proportional to the likelihood and no prior factor enters. The region $\mathcal{B}$ is covered by a square grid of $S = G^2$ single-emitter nodes $\xi_1, \dots, \xi_S$ of spacing Δ_g , where $G = 3\Delta_x/\Delta_g$. The posterior is the joint over the two labeled positions, so it is supported on the ordered pairs (ξ_a, ξ_b) , of which there are S^2 . For a pair, the mean photon count of pixel $\mathbf{k}$ is

$$v(\mathbf{k}; \xi_a, \xi_b) = s(\mathbf{k}; \xi_a) + s(\mathbf{k}; \xi_b) + b(\mathbf{k}), \quad (11)$$

the log posterior of the pair equals the log-likelihood (3) up to an additive constant,

$$L_{a,b} = \sum_{\mathbf{k} \in \Omega} [V(\mathbf{k}) \ln v(\mathbf{k}; \xi_a, \xi_b) - v(\mathbf{k}; \xi_a, \xi_b)], \quad (12)$$

and the normalized posterior weight of the pair is

$$P_{a,b} = \frac{e^{L_{a,b}}}{\sum_{a'=1}^S \sum_{b'=1}^S e^{L_{a',b'}}}, \quad (13)$$

which sums to one over the S^2 ordered pairs and is exchangeable, $P_{a,b} = P_{b,a}$, by the symmetry of (11) in the two nodes. The labeled MMSE estimator, the posterior mean of each index, has its two estimates coincide because the two marginals are identical, the degeneracy of [preprint, Prop. 1]; ViT-PF avoids this degeneracy not by matching but by never forming labeled indices.

4.2 The matched Bayes estimator

The matched Bayes estimator minimizes the posterior mean of the matched loss L_M . On the grid the objective of a candidate pair $(\mathbf{y}_1, \mathbf{y}_2)$ is

$$J(\mathbf{y}_1, \mathbf{y}_2) = \sum_{a=1}^S \sum_{b=1}^S P_{a,b} \min \left\{ \|\mathbf{y}_i - \boldsymbol{\xi}_a\|^2 + \|\mathbf{y}_j - \boldsymbol{\xi}_b\|^2, i, j = 1, 2, i \neq j \right\}, \quad (14)$$

and the estimator is its minimizer $\hat{\boldsymbol{\theta}}_B = \arg \min_{\mathbf{y}_1, \mathbf{y}_2} J(\mathbf{y}_1, \mathbf{y}_2)$. The minimizer is found by a Lloyd iteration that alternates an assignment step and an update step. The assignment step matches every pair to the candidate by the smaller of the two terms of (14) through the indicator

$$I_{a,b} = \mathbf{1} \{ \|\mathbf{y}_1 - \boldsymbol{\xi}_a\|^2 + \|\mathbf{y}_2 - \boldsymbol{\xi}_b\|^2 \leq \|\mathbf{y}_1 - \boldsymbol{\xi}_b\|^2 + \|\mathbf{y}_2 - \boldsymbol{\xi}_a\|^2 \}, \quad (15)$$

where $I_{a,b} = 1$ selects the identity match and $I_{a,b} = 0$ the swap, and the update step sets each candidate to the posterior weighted mean of the nodes assigned to it,

$$\mathbf{y}_1 \leftarrow \sum_{a,b} P_{a,b} [I_{a,b} \boldsymbol{\xi}_a + (1 - I_{a,b}) \boldsymbol{\xi}_b], \quad \mathbf{y}_2 \leftarrow \sum_{a,b} P_{a,b} [I_{a,b} \boldsymbol{\xi}_b + (1 - I_{a,b}) \boldsymbol{\xi}_a]. \quad (16)$$

The two steps do not increase J and converge to a fixed point; to avoid a local minimizer the iteration is started from the pair of largest posterior weight and from several random pairs, and the fixed point of smallest J is kept as $\hat{\boldsymbol{\theta}}_B$. Because J equals the posterior mean of MD_H^2 for $M = 2$, $\hat{\boldsymbol{\theta}}_B$ minimizes the reported metric by construction. The iteration uses the posterior and hence the true system parameters, so $\hat{\boldsymbol{\theta}}_B$ uses the full system model at every frame, whereas the networks of Section 3 approximate the same function of the frame in a single forward pass.

4.3 Grid resolution and Monte Carlo error

The computation is exact up to two controlled errors. The first is the grid spacing Δ_g , whose quantization standard deviation is $\sigma_g = \Delta_g / \sqrt{12}$ by the uniform distribution of a position in a cell; a spacing of $\Delta_g = 5$ to 6 nm gives σ_g below 2 nm, far smaller than the estimator errors compared in Section 5. The second is the Monte Carlo average over the frames of one configuration; with N frames the relative standard error of the reported ARMSMD-H is near 5 percent at $N = 200$ and falls as $N^{-1/2}$, so a dataset of $N = 1000$ frames reduces it to near 2 percent. The same grid resolution is used for the sub-pixel classification map of ViT-PF only insofar as the two share the node construction; the classification grid spacing $\Delta_c = \Delta_x / r$ is a design

choice of the network reported in Section 5 and is distinct from the quadrature spacing Δ_g used to compute the benchmark.

Draft ends at Section 4. Sections 5 (results), 6 (discussion), and 7 (conclusion) to follow.